Aciclovir Population Parallel Workflow Report

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1 Sensitivity Analysis

Figure 1-1: Most sensitive parameters for AUC_tEnd of Aciclovir Plasma for individuals at percentiles 5, 50, 95 for Aciclovir Children, Aciclovir Adults.

Figure 1-2: Most sensitive parameters for Total body clearance/F of Aciclovir Plasma for individuals at percentiles 5, 50, 95 for Aciclovir Adults, Aciclovir Children.

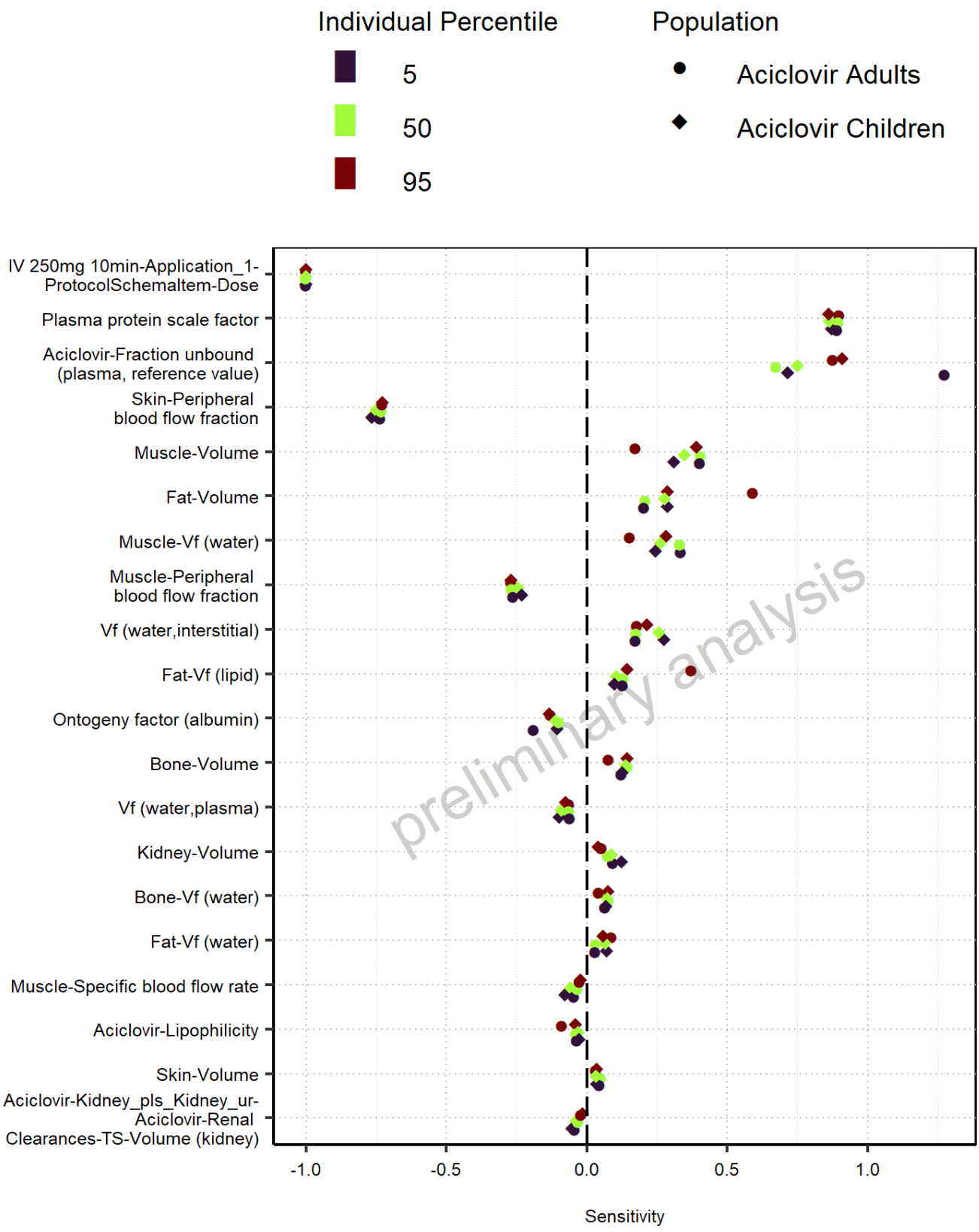

Figure 1-3: Most sensitive parameters for Vss (plasma)/F of Aciclovir Plasma for individuals at percentiles 5, 50, 95 for Aciclovir Adults, Aciclovir Children.

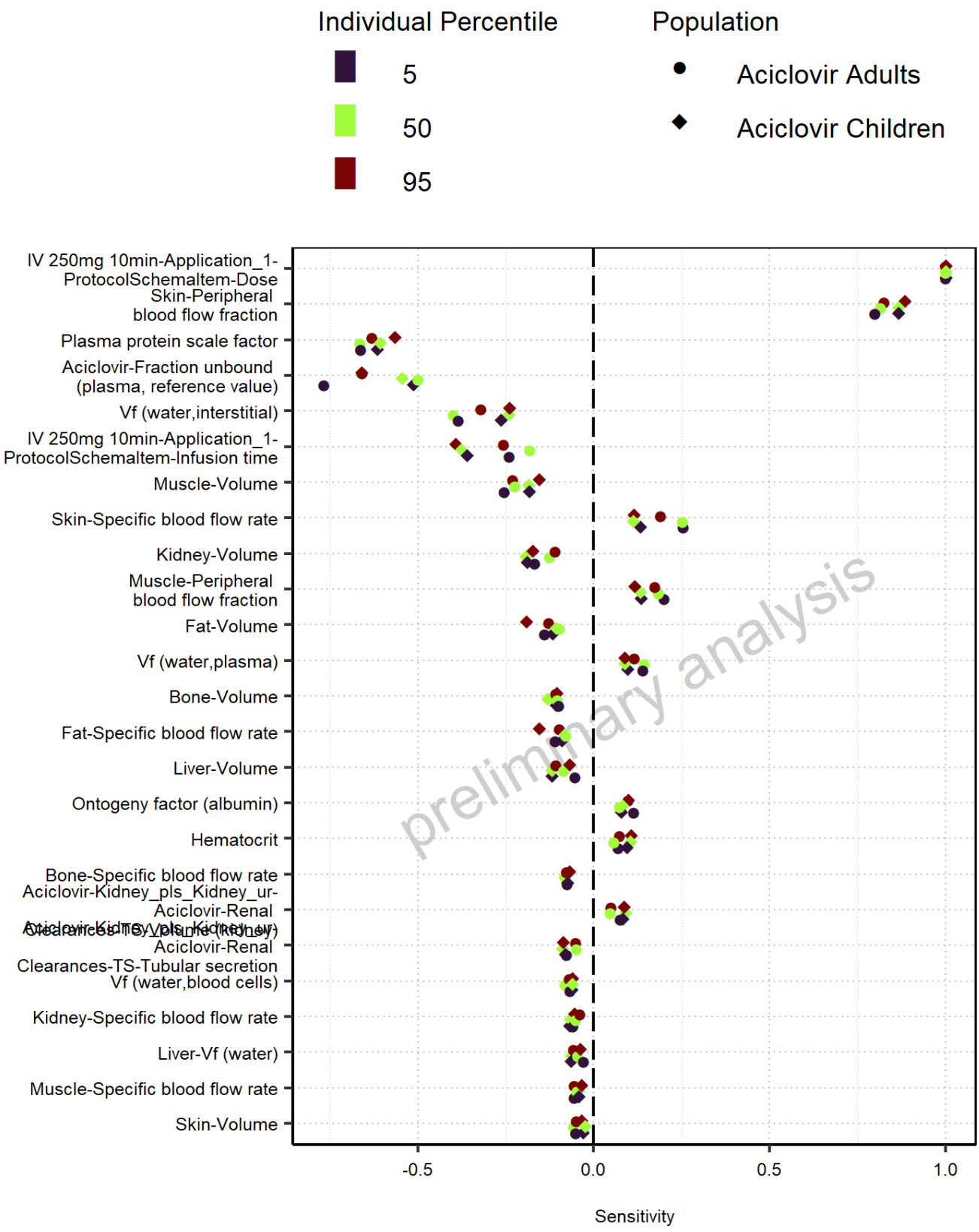

Figure 1-4: Most sensitive parameters for C_max of Aciclovir Plasma for individuals at percentiles 5, 50, 95 for Aciclovir Children, Aciclovir Adults.

2 PK parameters

2.1 PK Parameters of Aciclovir Plasma

2.1.1 AUC_tEnd

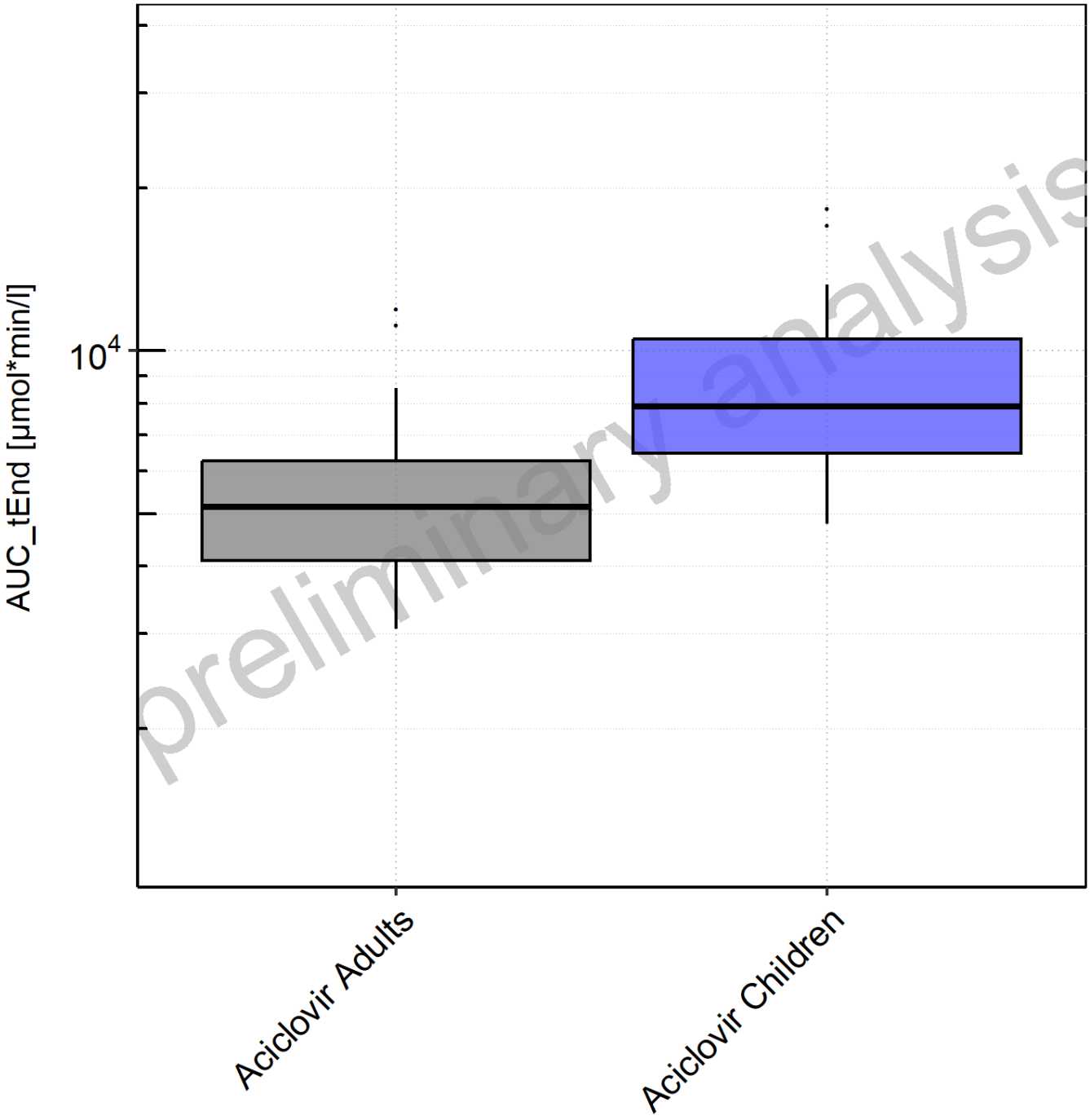

Figure 2-1: AUC_tEnd of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children shown as box-whisker plot, which indicates the 5th, 25th, 50th, 75th, 95th percentiles in logarithmic scale.

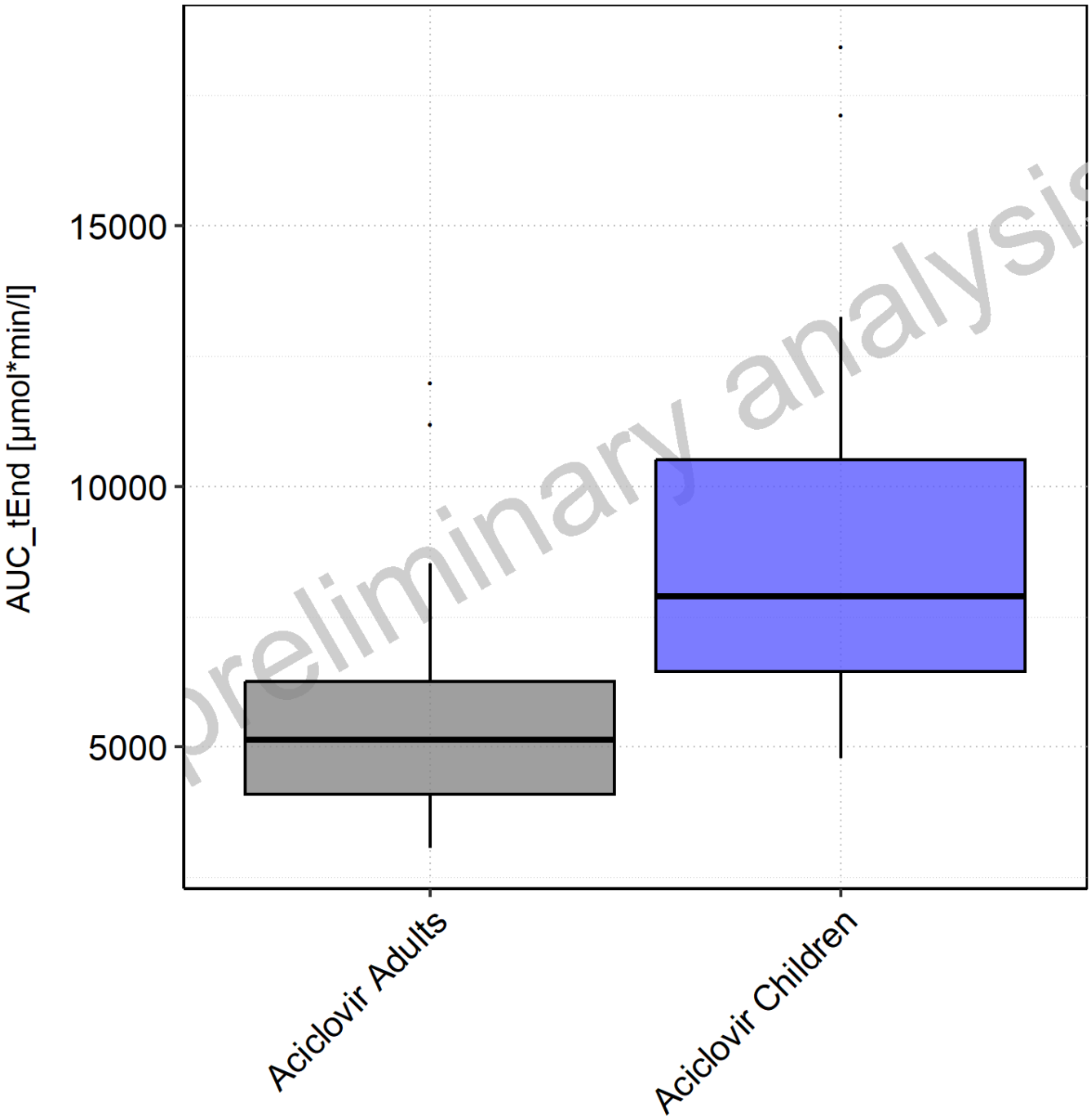

Figure 2-2: AUC_tEnd of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children shown as box-whisker plot, which indicates the 5th, 25th, 50th, 75th, 95th percentiles in linear scale.

Table 2-1: Population statistics summarizing AUC_tEnd of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children reported in [µmol*min/l]

Population	N	5th pctl	25th pctl	50th pctl	75th pctl	95th pctl	Mean	SD	Geo Mean	Geo SD
Aciclovir Adults	100	3059.31	4093.57	5140.20	6258.52	8520.73	5387.42	1792.62	5117.26	1.38
Aciclovir Children	100	4780.92	6455.52	7889.49	10507.43	13250.47	8594.20	2965.27	8135.55	1.39

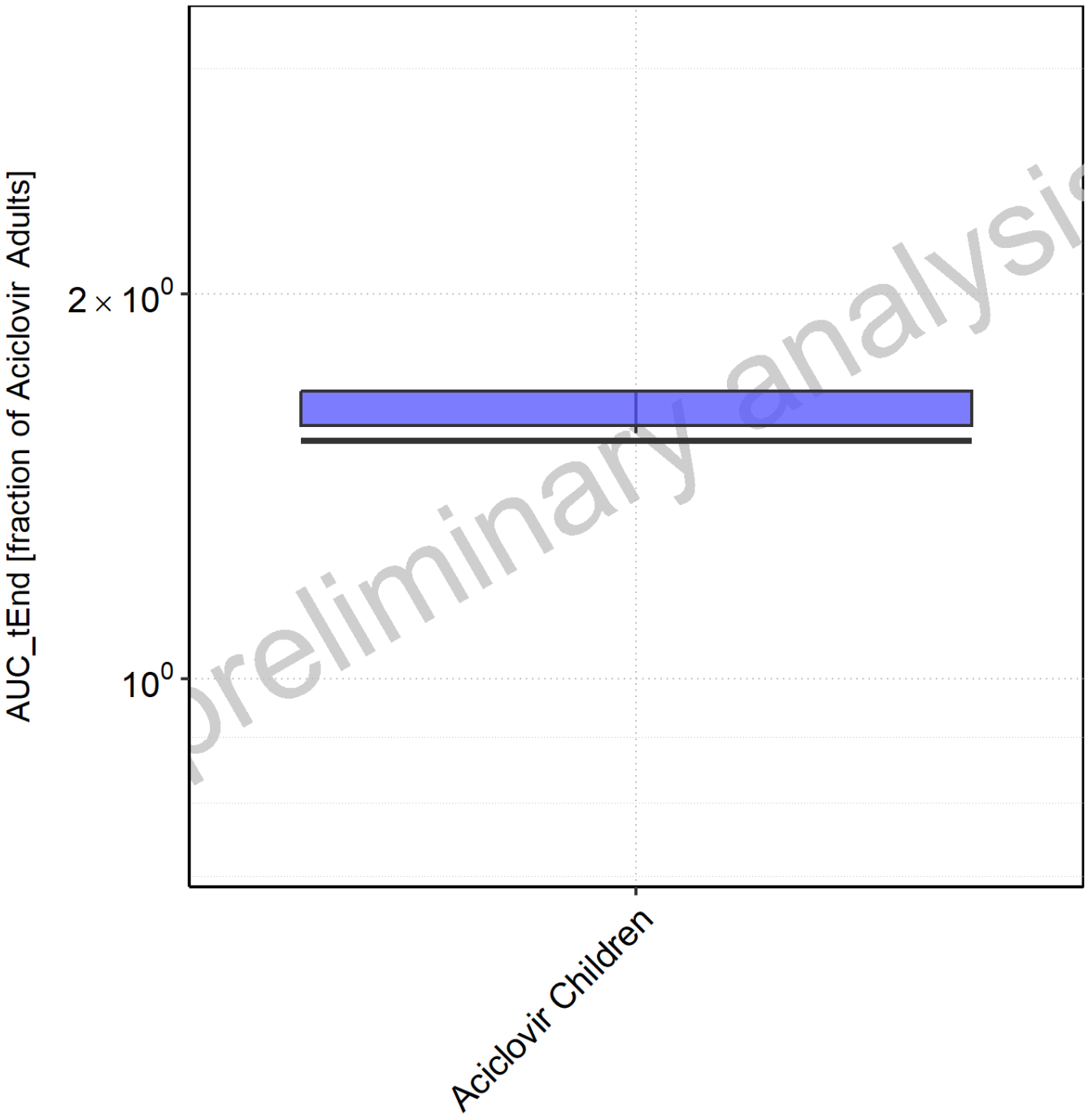

Figure 2-3: Ratio of population summary statistics of AUC_tEnd of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults shown as box-whisker plot, which indicates ratios of the 5th, 25th, 50th, 75th, 95th percentiles in logarithmic scale.

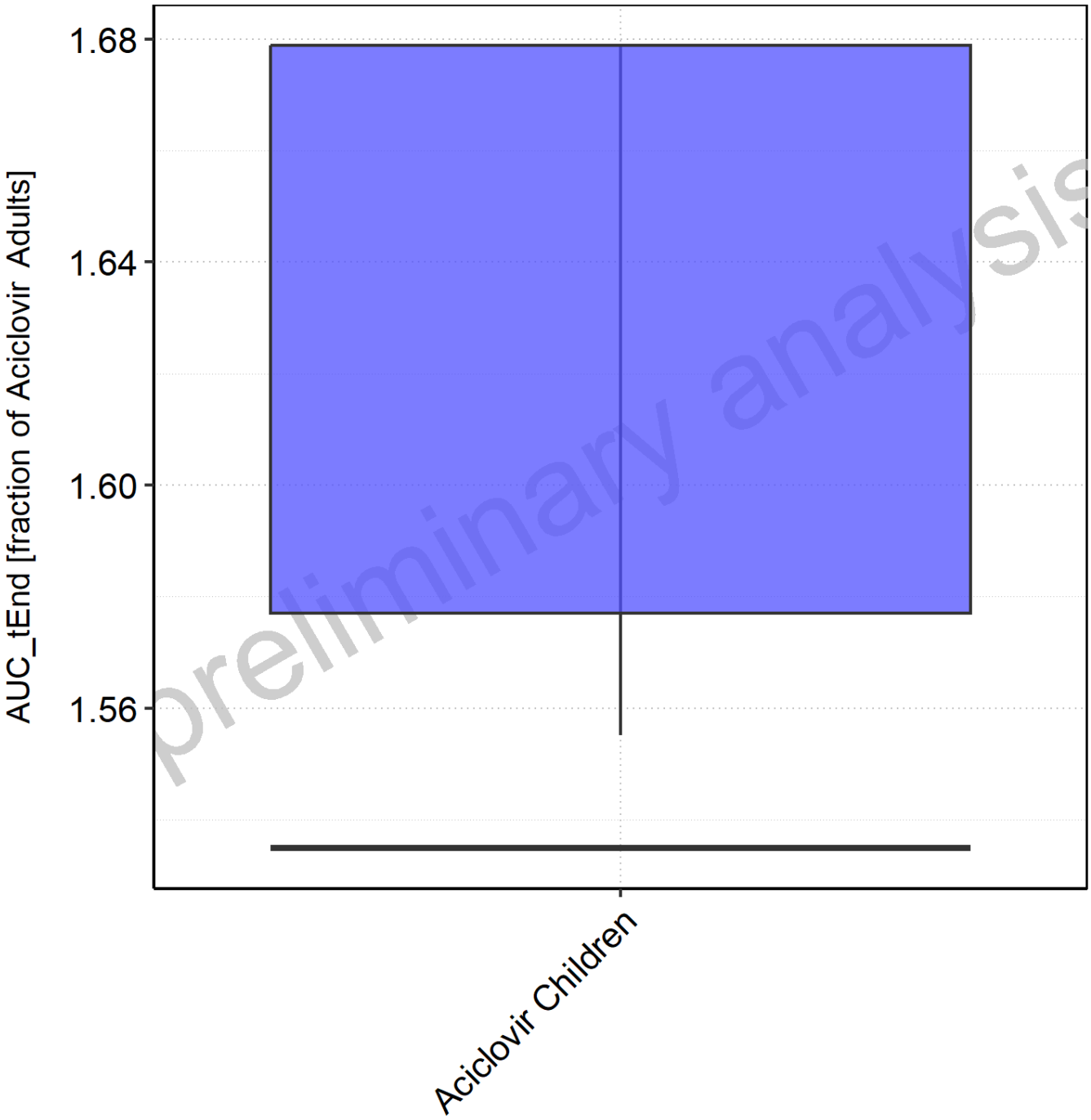

Figure 2-4: Ratio of population summary statistics of AUC_tEnd of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults shown as box-whisker plot, which indicates ratios of the 5th, 25th, 50th, 75th, 95th percentiles in linear scale.

Table 2-2: Ratio of population summary statistics of AUC_tEnd of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults

Population	N	5th pctl	25th pctl	50th pctl	75th pctl	95th pctl	Mean	SD	Geo Mean	Geo SD
Aciclovir Children	100	1.56	1.58	1.53	1.68	1.56	1.60	1.65	1.59	1.01

2.1.2 Total body clearance/F

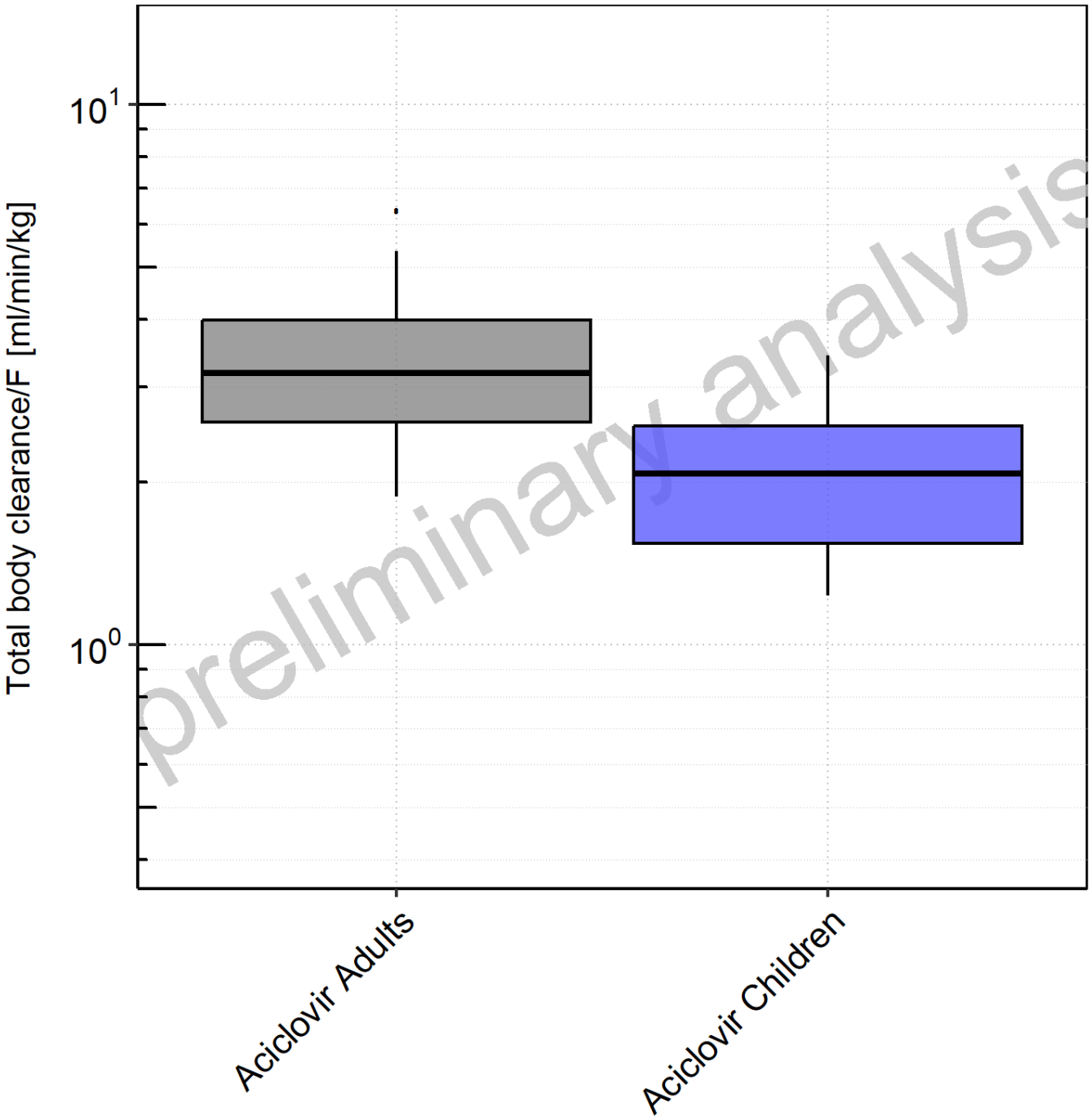

Figure 2-5: Total body clearance/F of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children shown as box-whisker plot, which indicates the 5th, 25th, 50th, 75th, 95th percentiles in logarithmic scale.

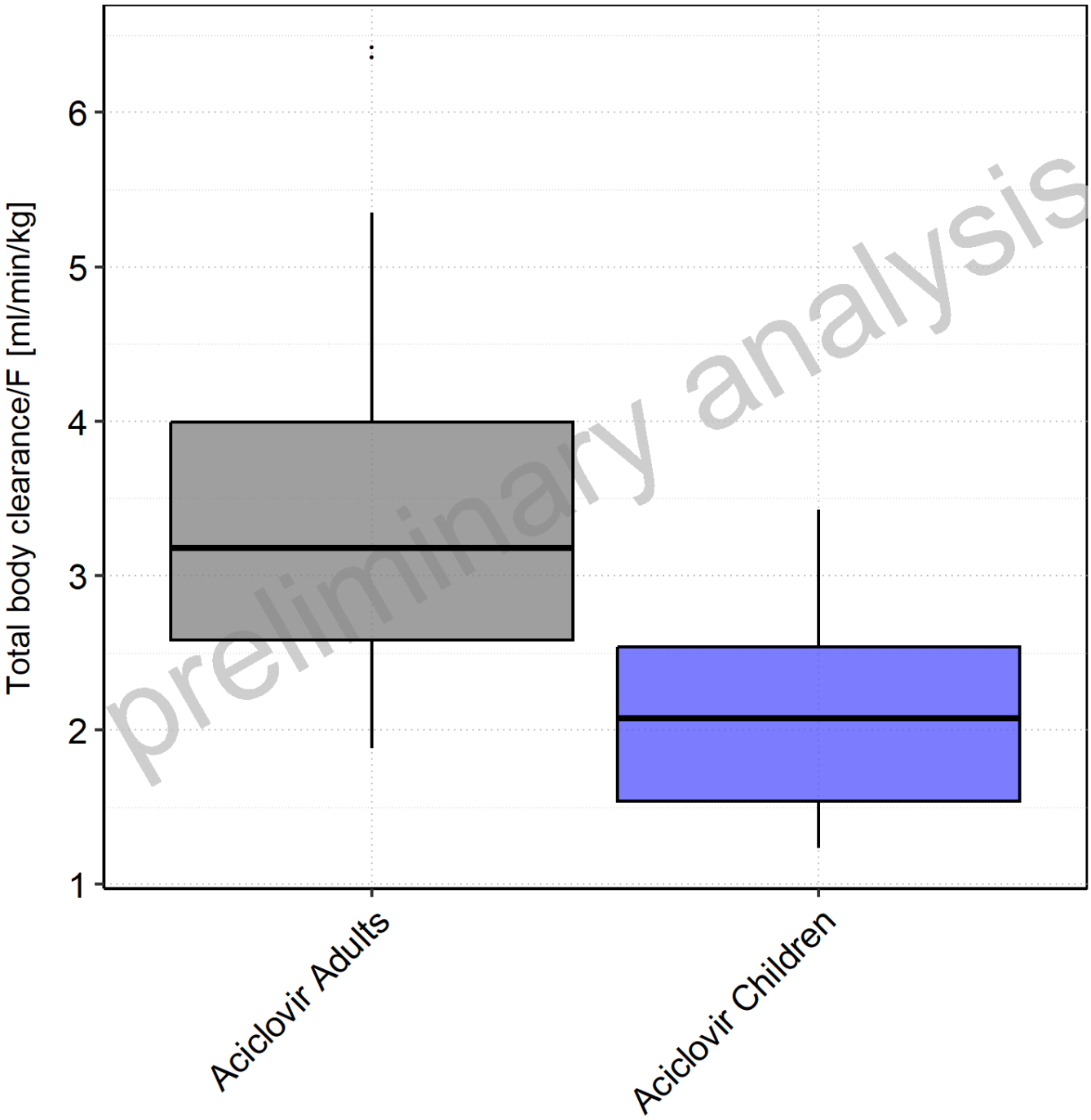

Figure 2-6: Total body clearance/F of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children shown as box-whisker plot, which indicates the 5th, 25th, 50th, 75th, 95th percentiles in linear scale.

Table 2-3: Population statistics summarizing Total body clearance/F of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children reported in [ml/min/kg]

Population	N	5th pctl	25th pctl	50th pctl	75th pctl	95th pctl	Mean	SD	Geo Mean	Geo SD
Aciclovir Adults	100	1.88	2.58	3.18	3.99	5.35	3.34	1.09	3.17	1.39
Aciclovir Children	100	1.23	1.54	2.08	2.54	3.43	2.12	0.68	2.01	1.39

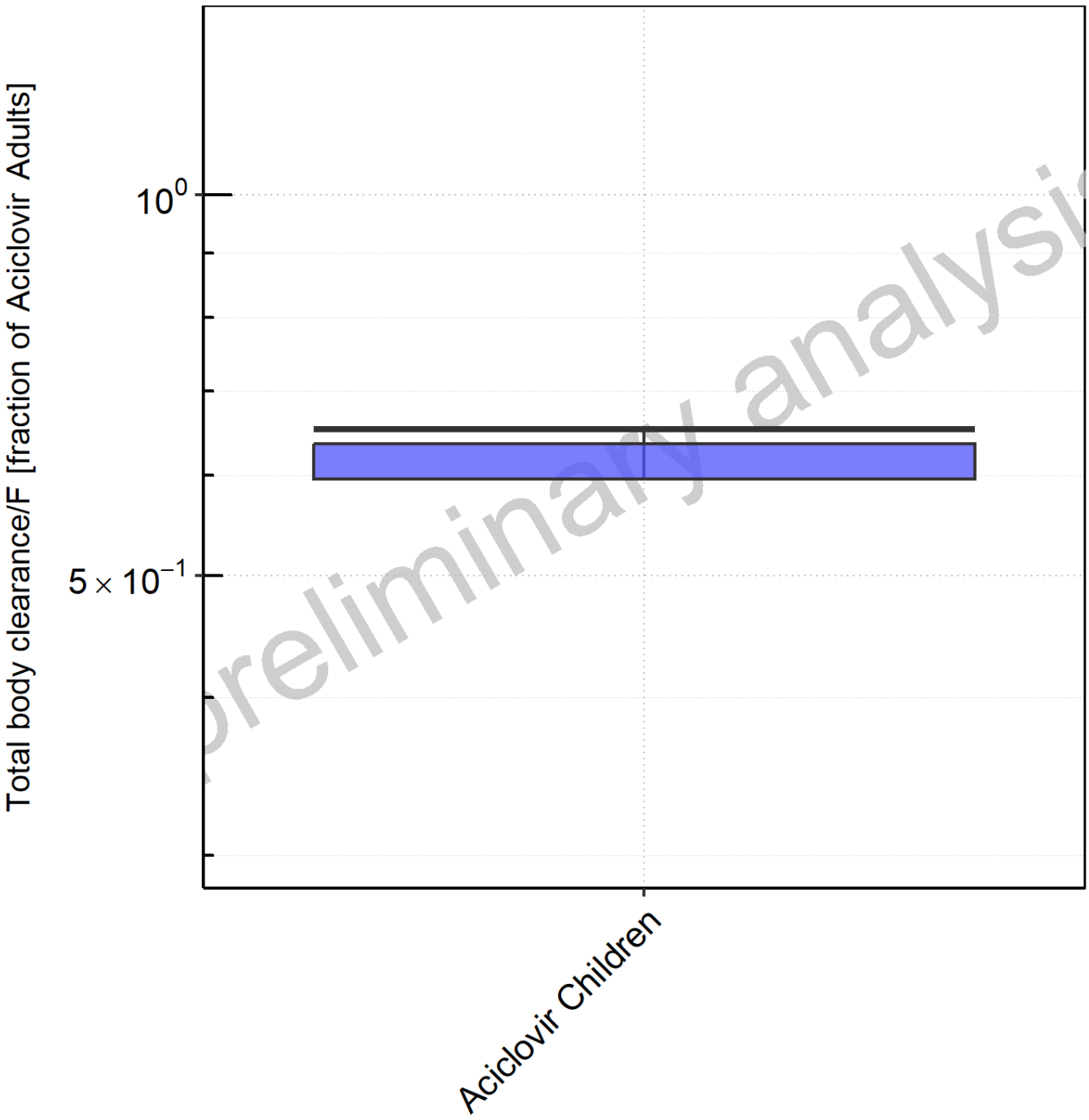

Figure 2-7: Ratio of population summary statistics of Total body clearance/F of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults shown as box-whisker plot, which indicates ratios of the 5th, 25th, 50th, 75th, 95th percentiles in logarithmic scale.

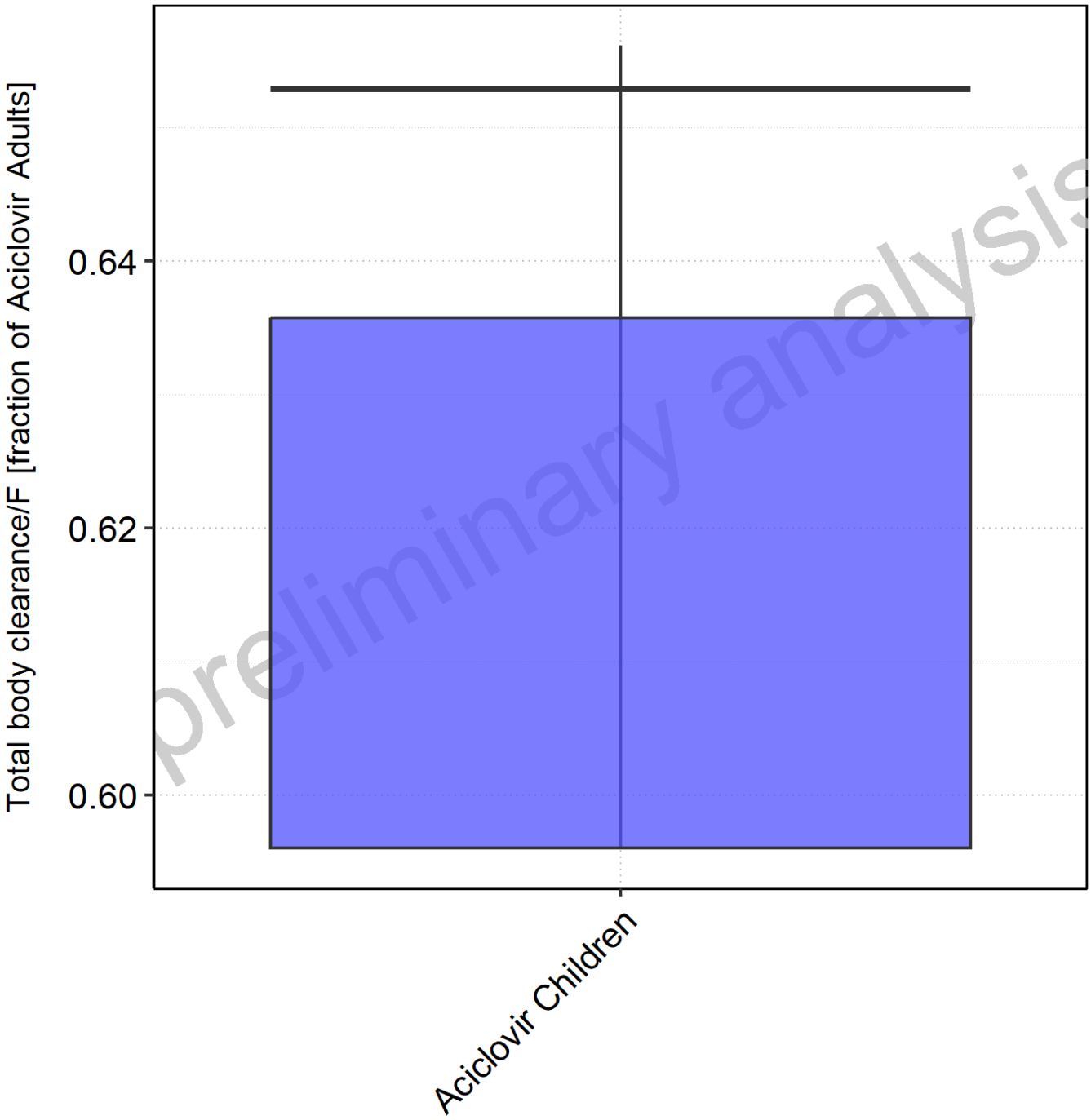

Figure 2-8: Ratio of population summary statistics of Total body clearance/F of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults shown as box-whisker plot, which indicates ratios of the 5th, 25th, 50th, 75th, 95th percentiles in linear scale.

Table 2-4: Ratio of population summary statistics of Total body clearance/F of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults

Population	N	5th pctl	25th pctl	50th pctl	75th pctl	95th pctl	Mean	SD	Geo Mean	Geo SD
Aciclovir Children	100	0.66	0.60	0.65	0.64	0.64	0.63	0.62	0.63	1.00

2.1.3 Vss (plasma)/F

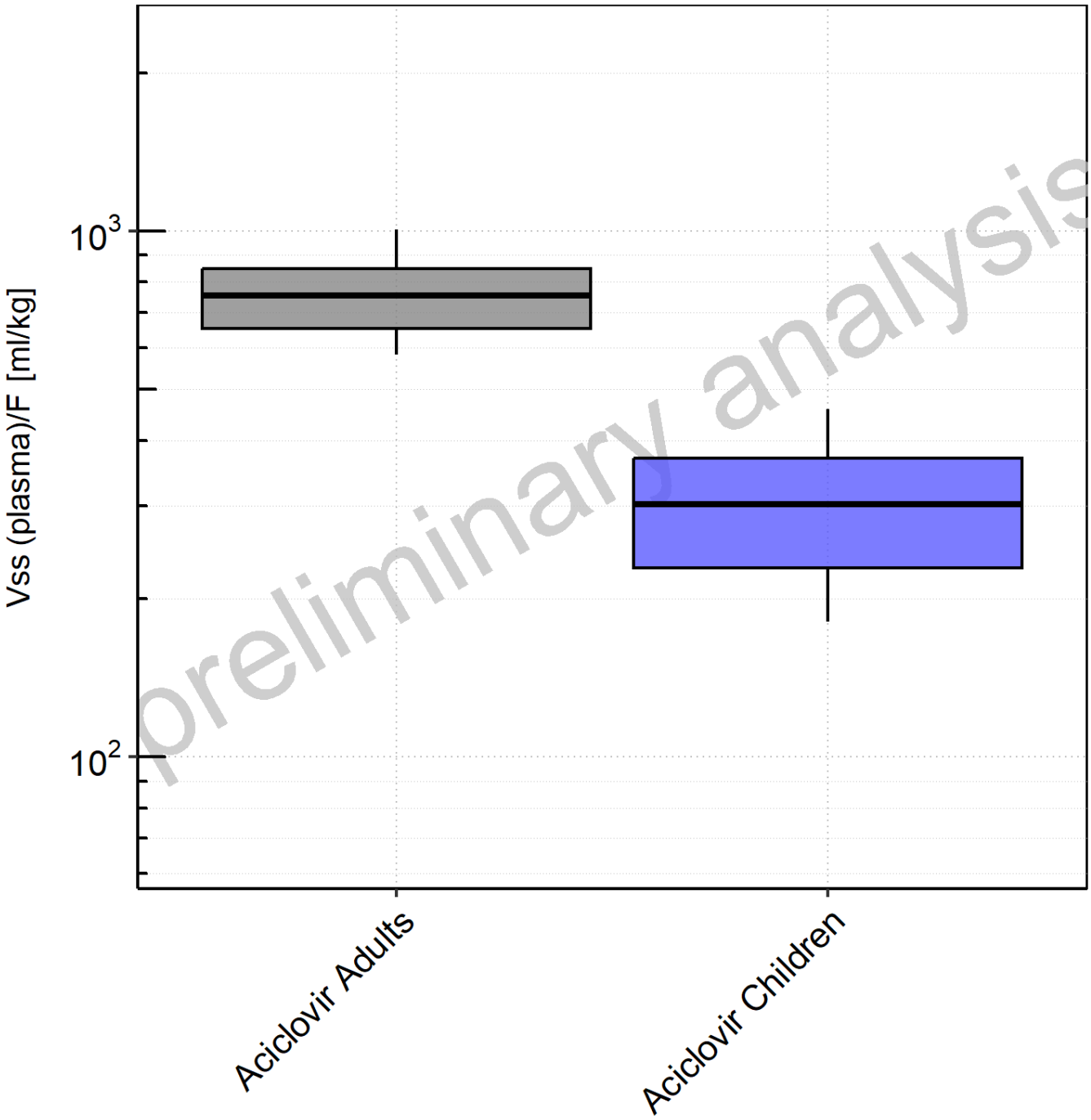

Figure 2-9: Vss (plasma)/F of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children shown as box-whisker plot, which indicates the 5th, 25th, 50th, 75th, 95th percentiles in logarithmic scale.

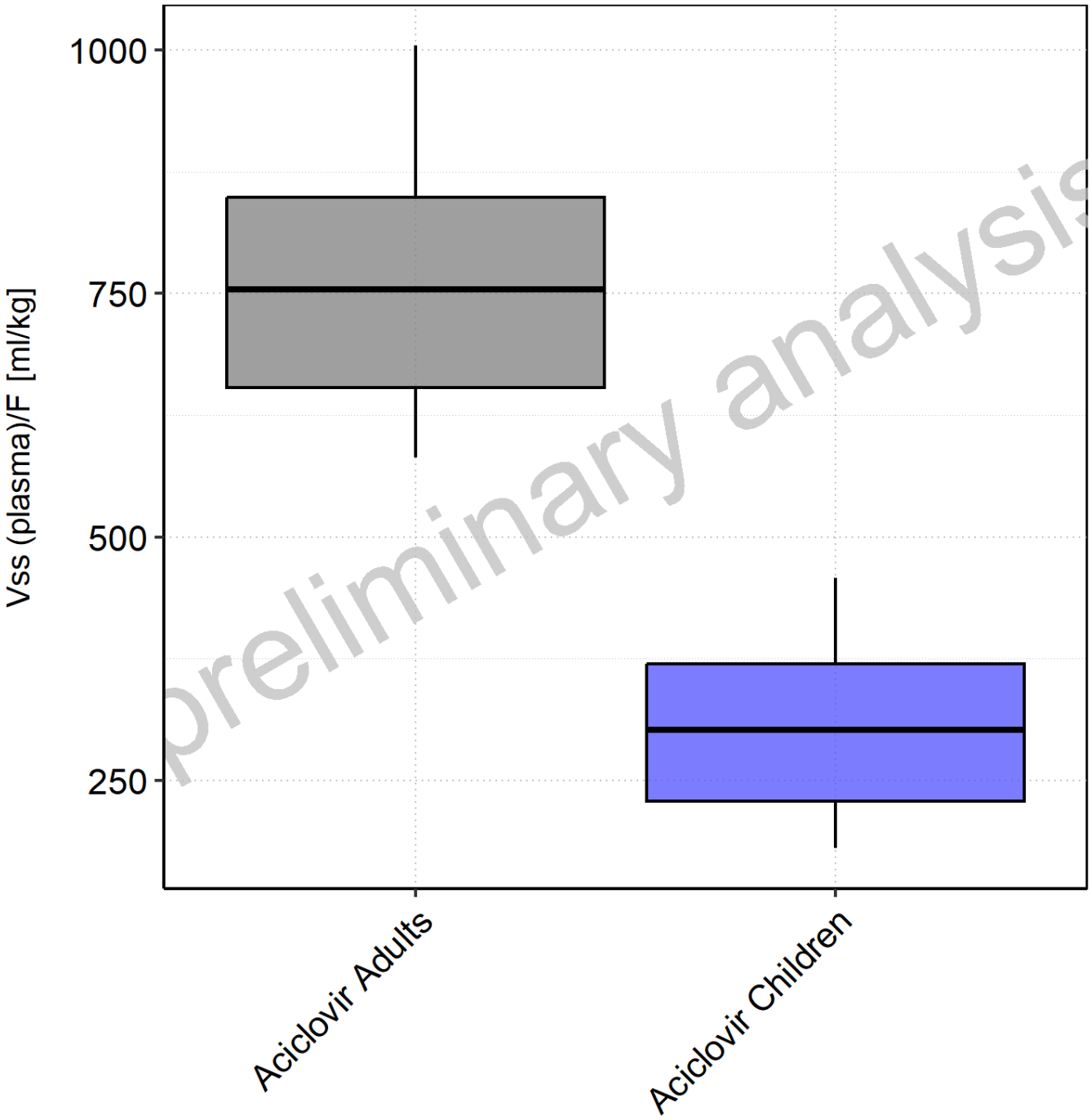

Figure 2-10: Vss (plasma)/F of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children shown as box-whisker plot, which indicates the 5th, 25th, 50th, 75th, 95th percentiles in linear scale.

Table 2-5: Population statistics summarizing Vss (plasma)/F of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children reported in [ml/kg]

Population	N	5th pctl	25th pctl	50th pctl	75th pctl	95th pctl	Mean	SD	Geo Mean	Geo SD
Aciclovir Adults	100	581.90	653.15	753.84	848.39	1004.80	760.07	135.65	748.29	1.19
Aciclovir Children	100	180.58	228.71	302.41	369.64	457.91	304.05	88.39	291.30	1.35

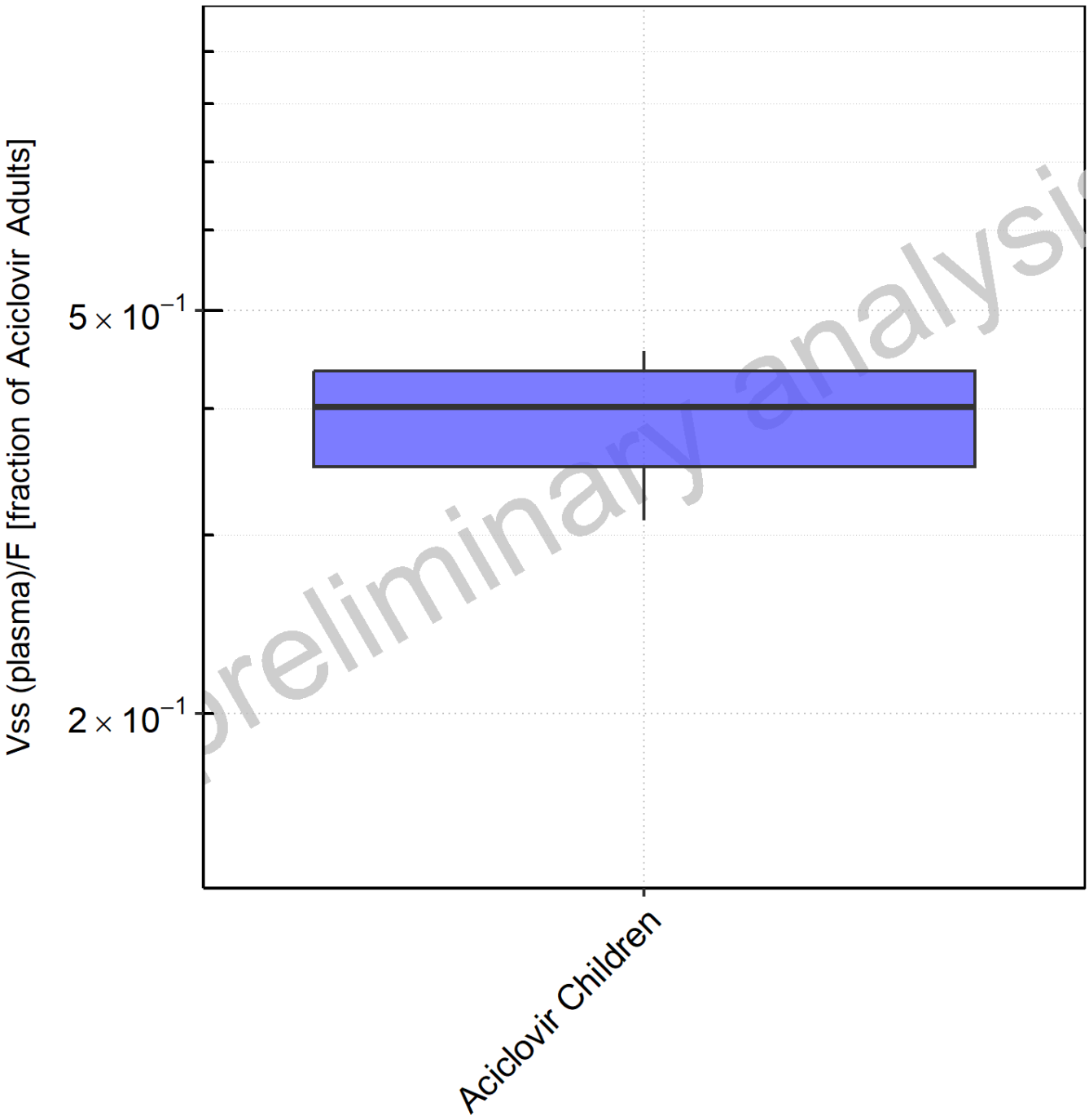

Figure 2-11: Ratio of population summary statistics of Vss (plasma)/F of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults shown as box-whisker plot, which indicates ratios of the 5th, 25th, 50th, 75th, 95th percentiles in logarithmic scale.

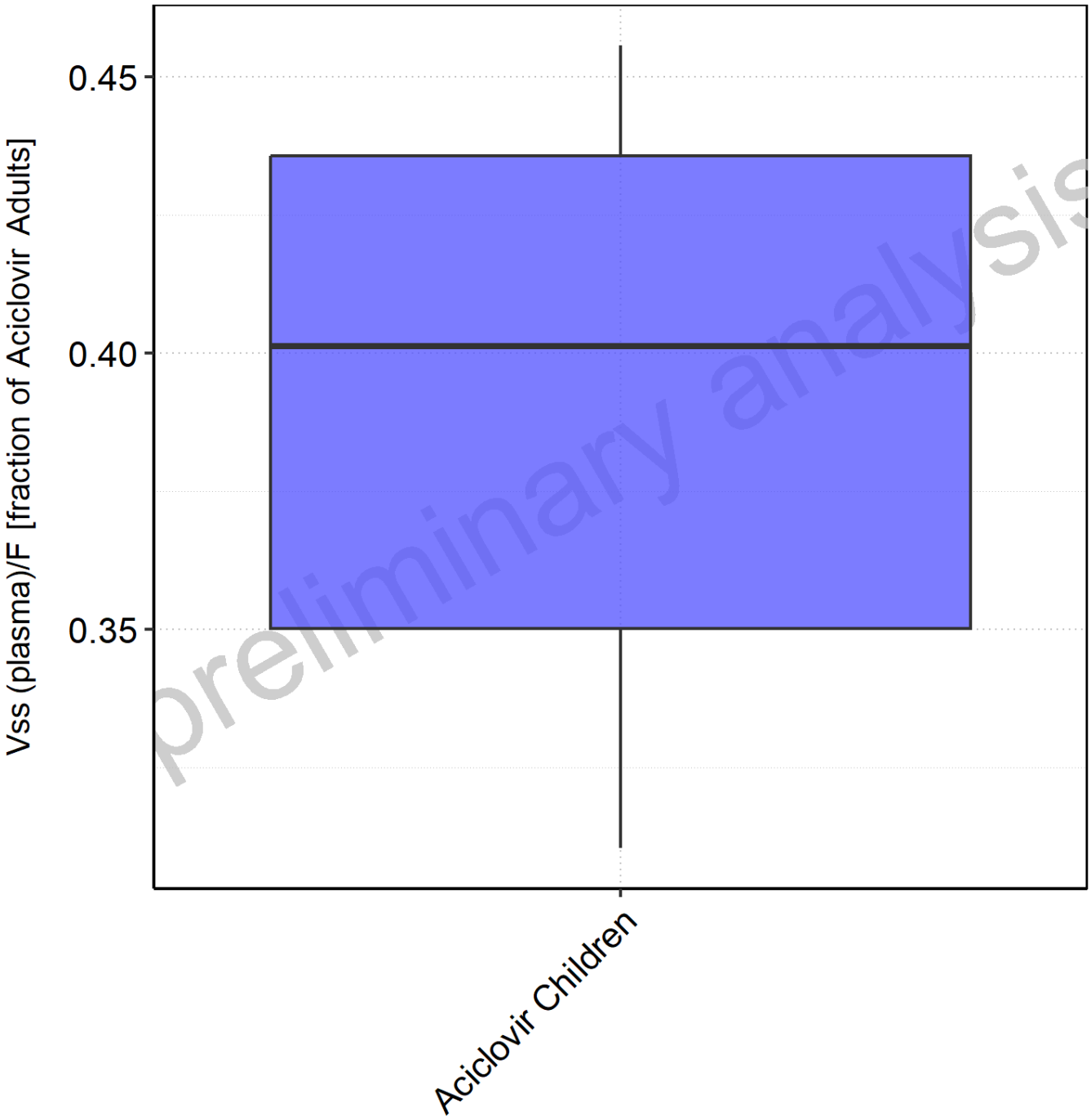

Figure 2-12: Ratio of population summary statistics of Vss (plasma)/F of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults shown as box-whisker plot, which indicates ratios of the 5th, 25th, 50th, 75th, 95th percentiles in linear scale.

Table 2-6: Ratio of population summary statistics of Vss (plasma)/F of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults

Population	N	5th pctl	25th pctl	50th pctl	75th pctl	95th pctl	Mean	SD	Geo Mean	Geo SD
Aciclovir Children	100	0.31	0.35	0.40	0.44	0.46	0.40	0.65	0.39	1.13

2.1.4 C_max

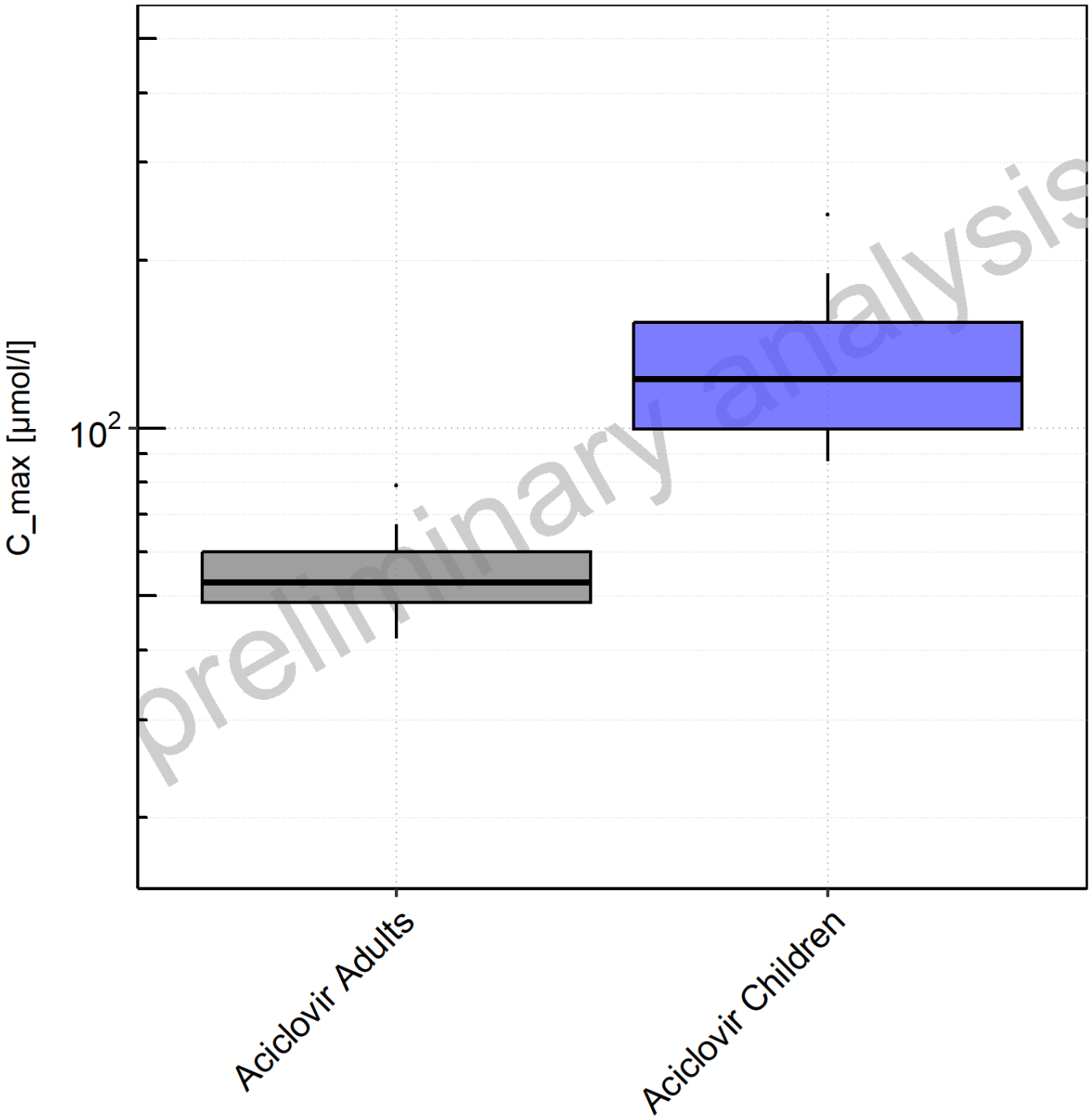

Figure 2-13: C_max of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children shown as box-whisker plot, which indicates the 5th, 25th, 50th, 75th, 95th percentiles in logarithmic scale.

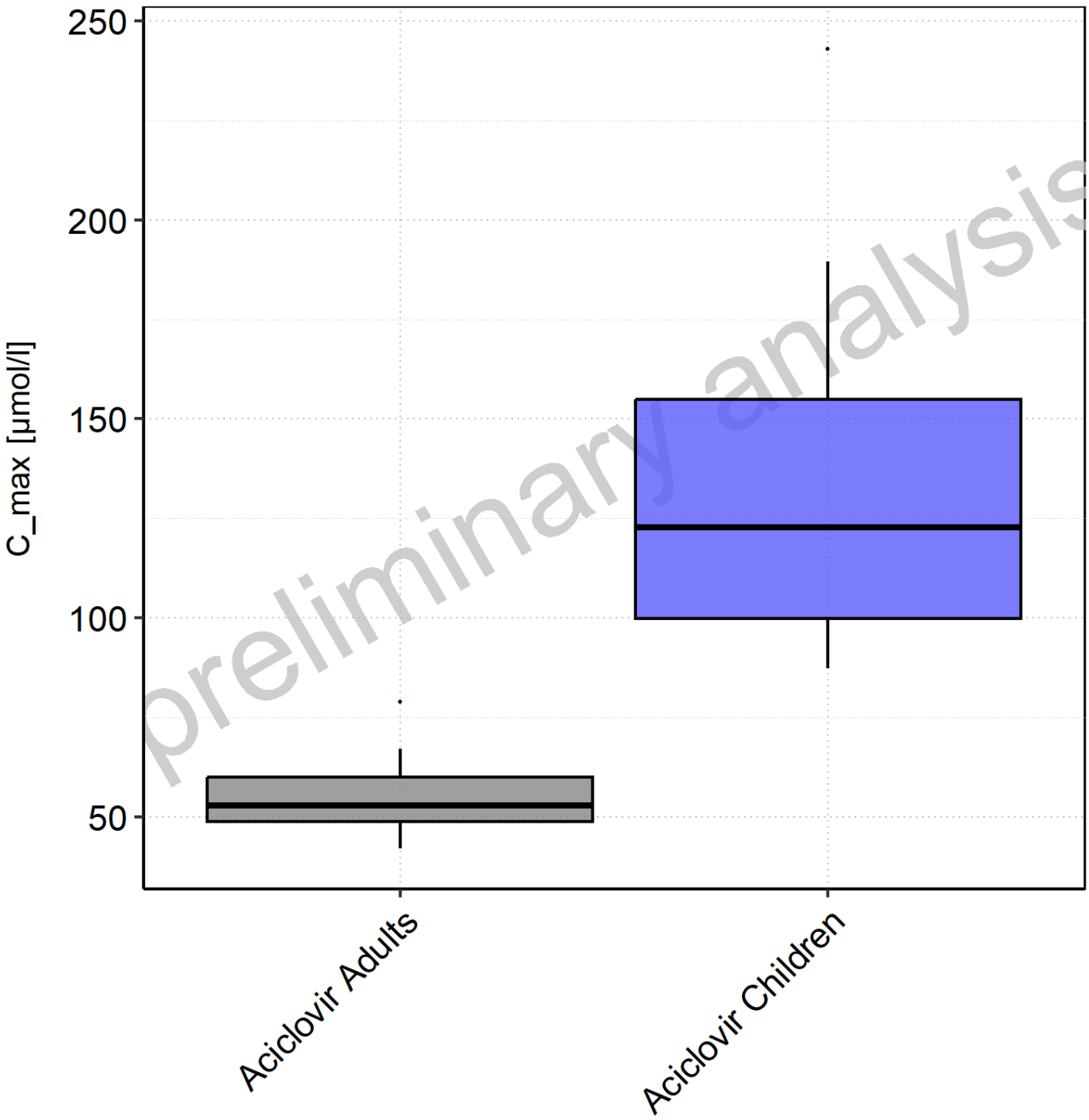

Figure 2-14: C_max of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children shown as box-whisker plot, which indicates the 5th, 25th, 50th, 75th, 95th percentiles in linear scale.

Table 2-7: Population statistics summarizing C_max of Aciclovir Plasma for Aciclovir Adults, Aciclovir Children reported in [µmol/l]

Population	N	5th pctl	25th pctl	50th pctl	75th pctl	95th pctl	Mean	SD	Geo Mean	Geo SD
Aciclovir Adults	100	41.99	48.73	52.90	59.91	67.16	54.30	8.02	53.72	1.16
Aciclovir Children	100	87.24	99.81	122.66	154.90	189.55	130.01	34.15	125.82	1.29

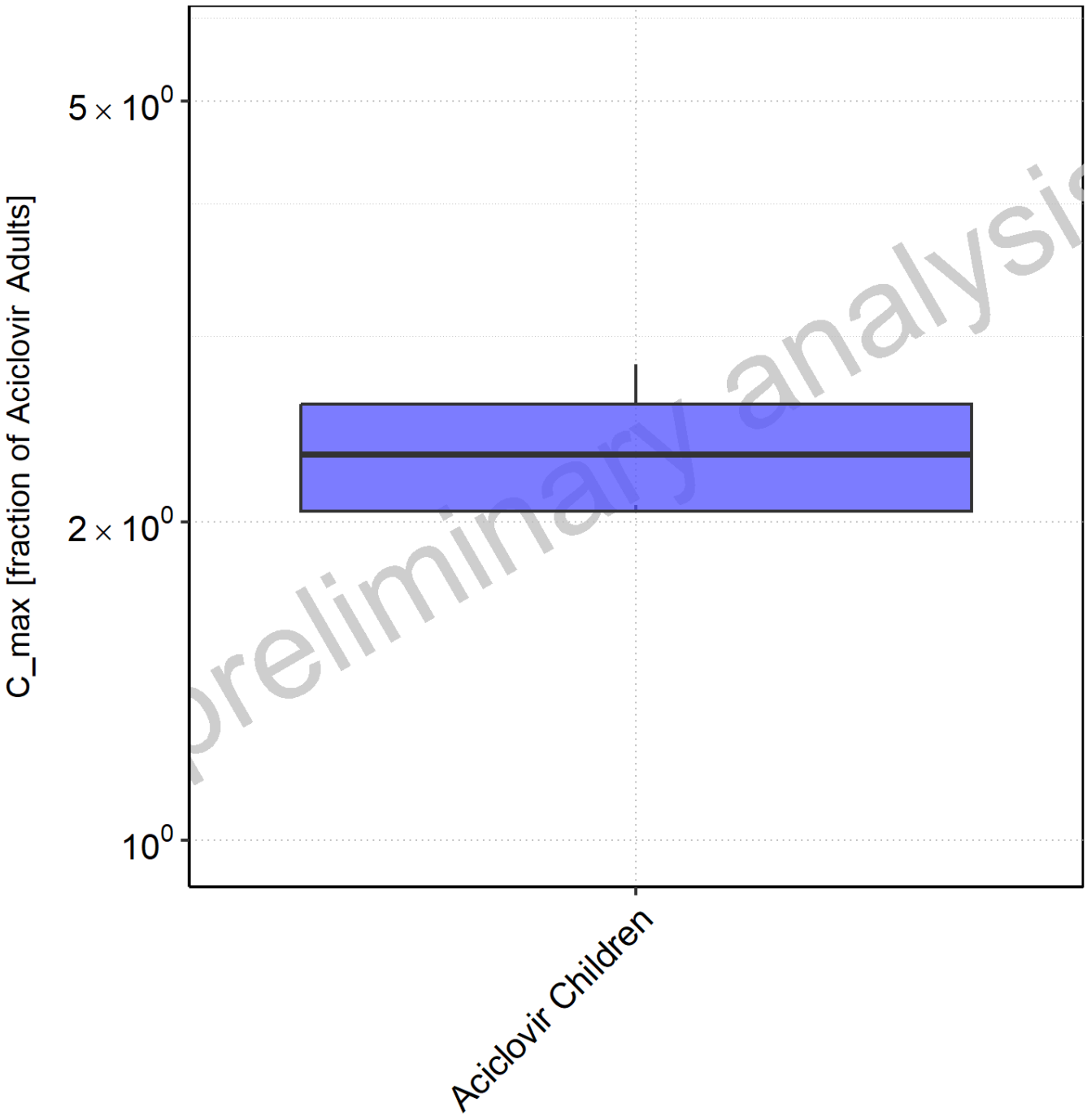

Figure 2-15: Ratio of population summary statistics of C_max of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults shown as box-whisker plot, which indicates ratios of the 5th, 25th, 50th, 75th, 95th percentiles in logarithmic scale.

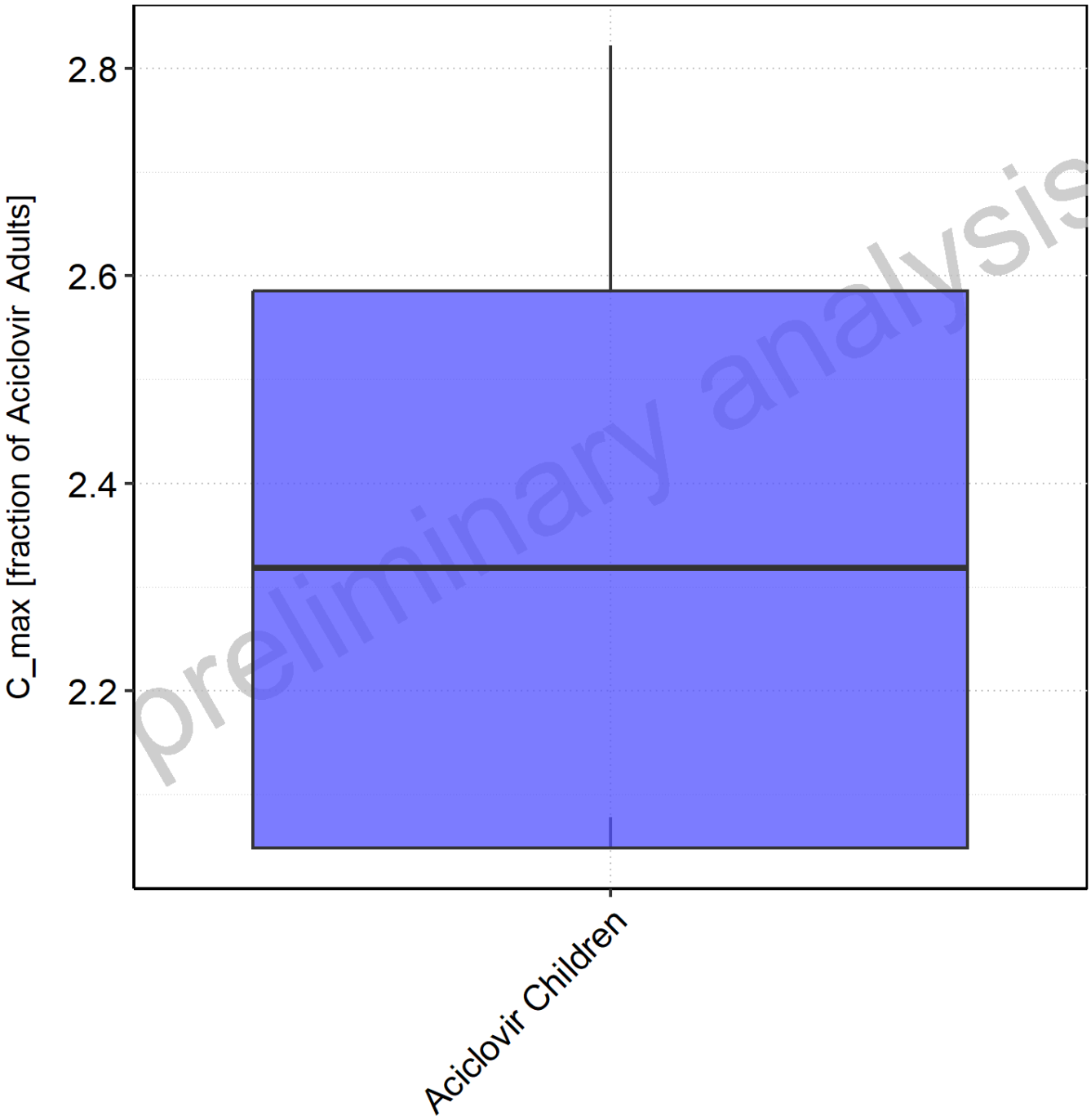

Figure 2-16: Ratio of population summary statistics of C_max of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults shown as box-whisker plot, which indicates ratios of the 5th, 25th, 50th, 75th, 95th percentiles in linear scale.

Table 2-8: Ratio of population summary statistics of C_max of Aciclovir Plasma for Aciclovir Children in comparison to Aciclovir Adults

Population	N	5th pctl	25th pctl	50th pctl	75th pctl	95th pctl	Mean	SD	Geo Mean	Geo SD
Aciclovir Children	100	2.08	2.05	2.32	2.59	2.82	2.39	4.26	2.34	1.11

3 Time profiles and residual plots

3.1 Time profiles and residual plots for Aciclovir Adults

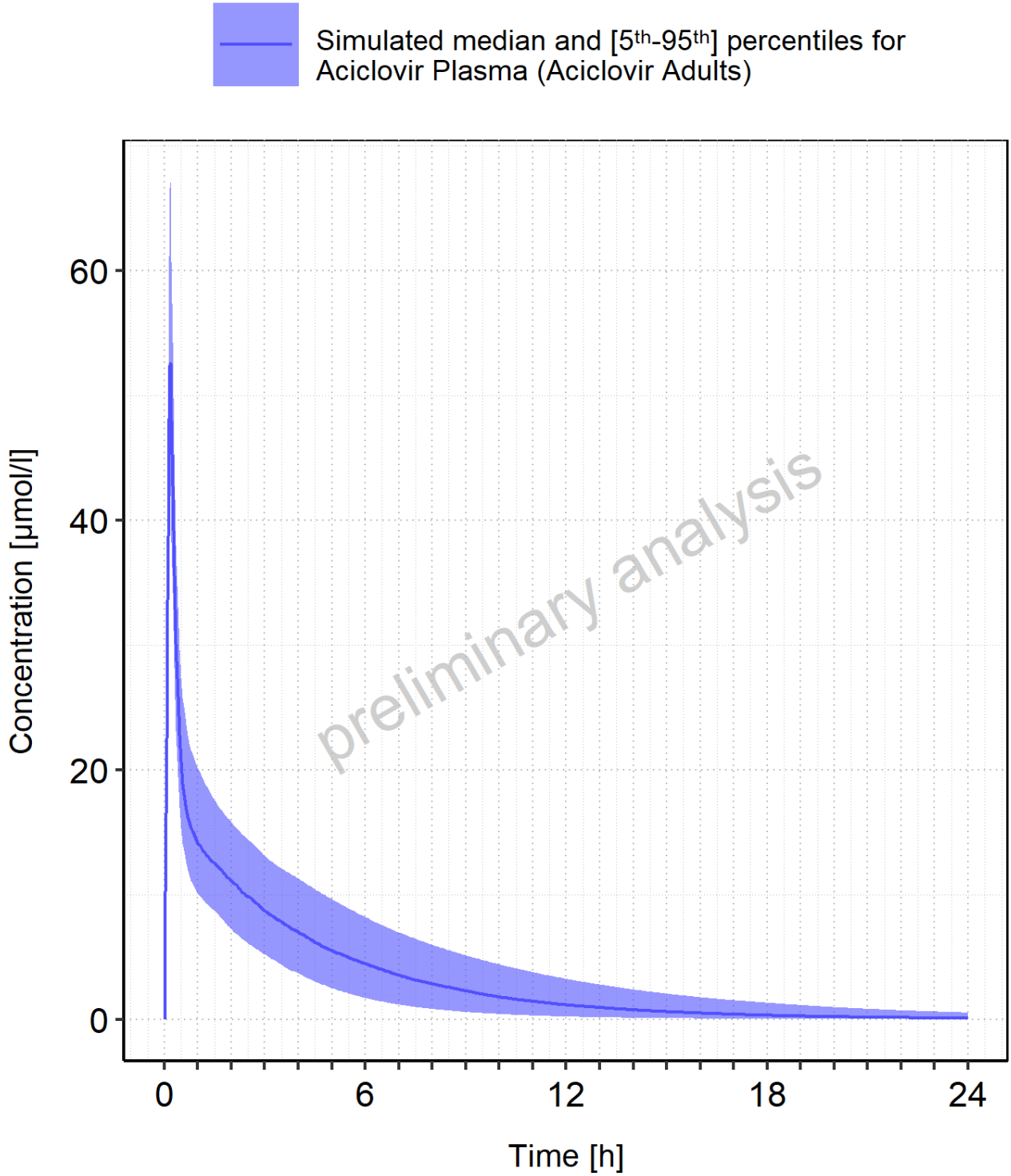

Figure 3-1: Time profiles for Aciclovir Adults. Time profiles are plotted in a linear scale.

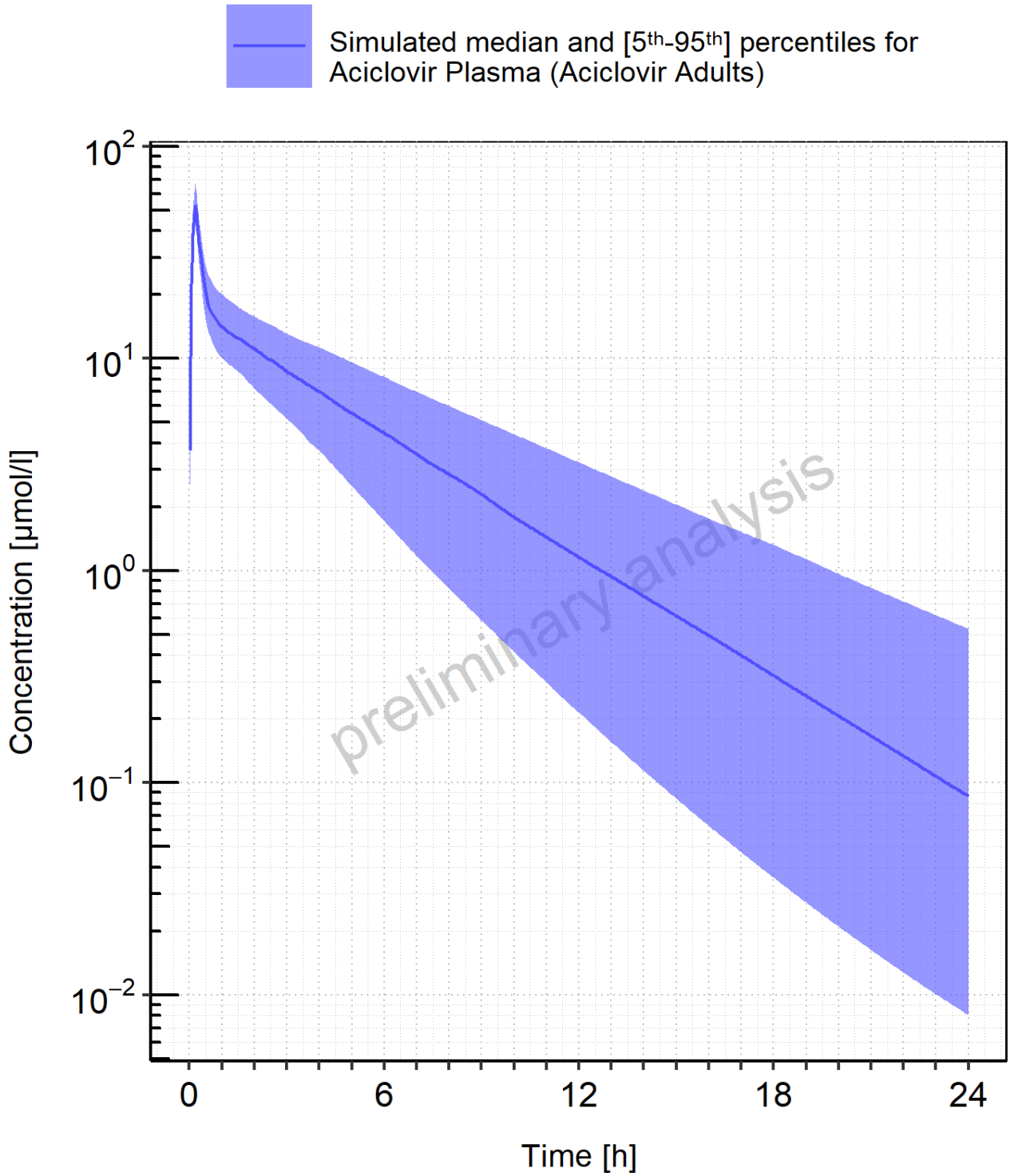

Figure 3-2: Time profiles for Aciclovir Adults. Time profiles are plotted in a logarithmic scale.

3.2 Time profiles and residual plots for Aciclovir Children

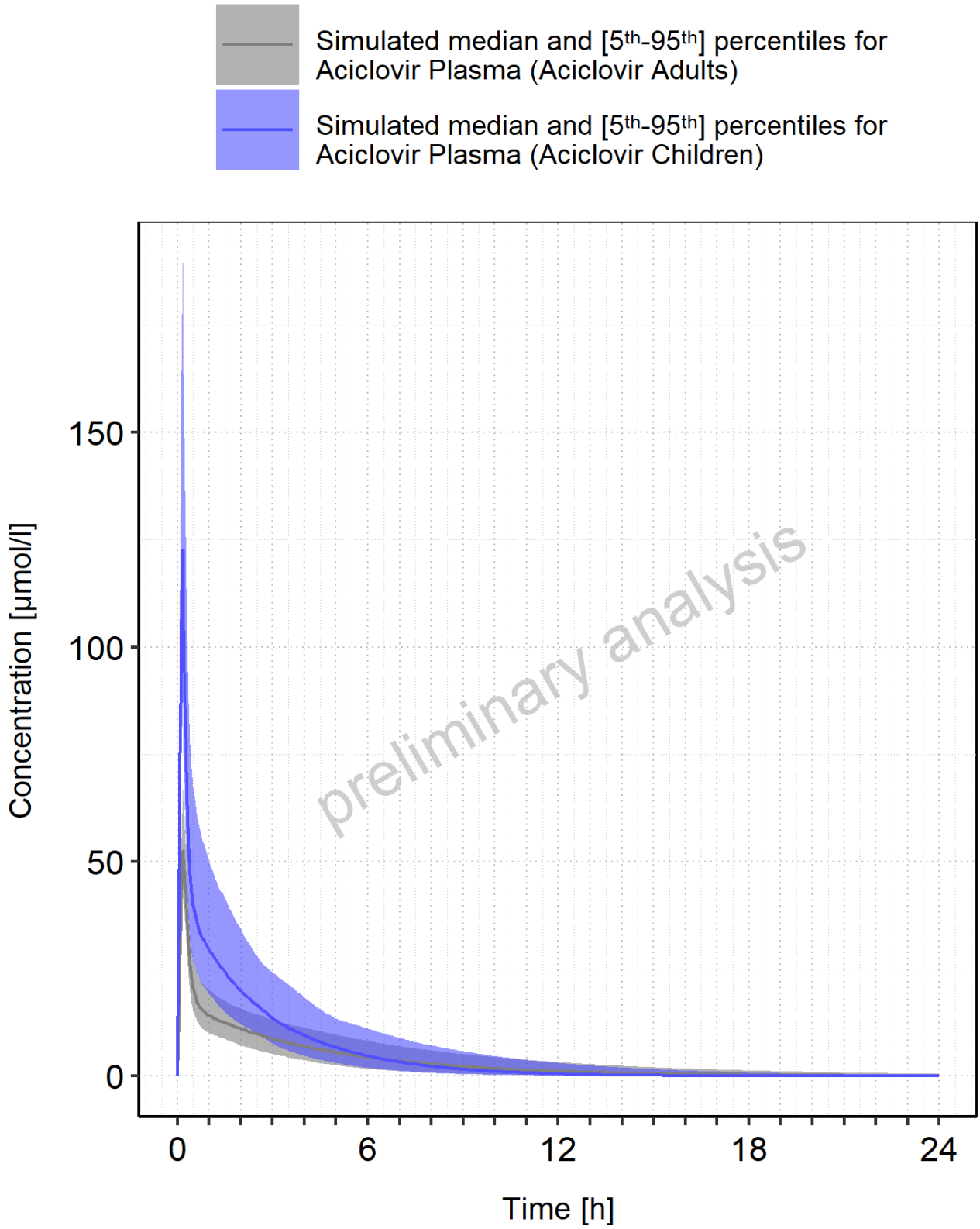

Figure 3-3: Time profiles for Aciclovir Children. Time profiles are plotted in a linear scale.

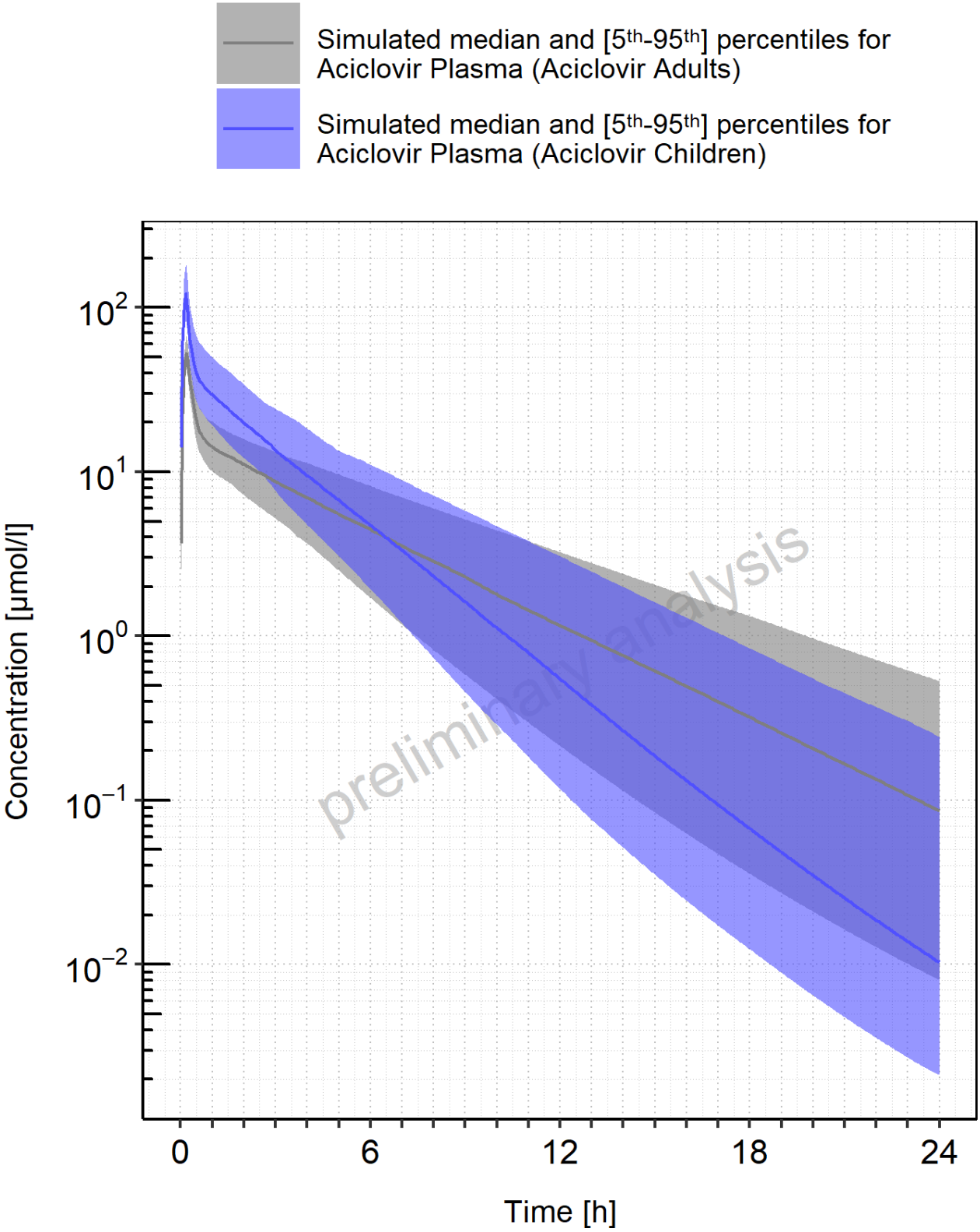

Figure 3-4: Time profiles for Aciclovir Children. Time profiles are plotted in a logarithmic scale.